$$J_0 \sim \mathcal{N}(0, N^{-1})$$

$$J_1 = (1-\alpha^2 N)^{\frac{1}{2}} J_0 + \alpha \xi - \eta \cdot \xi^T$$

$$\begin{pmatrix} \alpha \xi^T \\ \vdots \end{pmatrix}$$

$$J_p = (1-\alpha^2 N)^{\frac{p}{2}} J_0 + \sum_k (1-\alpha^2 N)^{\frac{k-1}{2}} \alpha \xi^k$$

$$P = CN$$

$$J_p \xi^q = \underbrace{(1-\alpha^2 N)^{\frac{p}{2}}}_{\substack{\downarrow \\ 1}} \overset{0.3}{O(1)} + \sum_{k \neq q} (1-\alpha^2 N)^{\frac{k-1}{2}} \alpha^2 N \, \alpha \cdot \sqrt{N} + (1-\alpha^2 N)^{\frac{q-1}{2}} \alpha \cdot N$$

$$(1 \le q \le P)$$

$$\alpha \sqrt{N} \sqrt{CN}$$

$$\alpha N \sqrt{C}$$

$$\xi_i S_i = \sum_j W_{ij} \xi_j + \eta_i$$

$$\dot{x} = \tanh(S_i) - x_i$$

$$m = \xi \tanh(S)$$

high stab

S_i

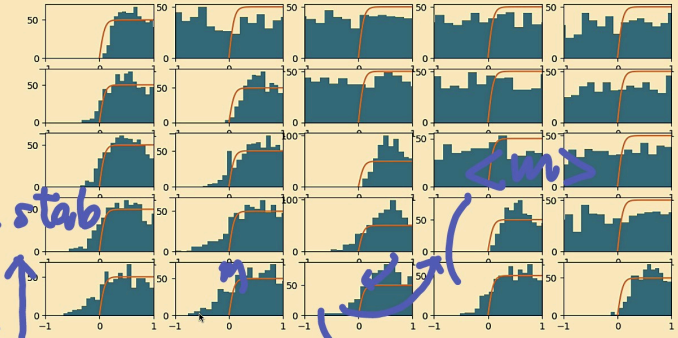
$\langle m \rangle$

η_i

```
In [22]:
#Dn=0.1
plt.figure(figsize=(12,6))
for i in range(5):
    for k in range(5):
        plt.subplot(5,5,5*k+i+1)

        J=np.copy(J_history[4*k]).reshape(N,N)
        tmp=J@xiset[:,4*i]+Iset[:,4*i]
        plt.hist(tmp*xiset[:,4*i],bins=50)
        plt.xlim(-1,5)
        print(i*4,k*4,np.sum(np.tanh(10*tmp)*xiset[:,4*i]))
```

```
0 0 995.8970607839267
0 4 921.6295810860518
0 8 891.6201800017877
0 12 841.3619418416579
0 16 787.0870924427808
4 0 2.2827781814411283
4 4 988.8806030302017
4 8 885.5395720608059
4 12 814.6103704567581
4 16 775.1085657309632
8 0 -8.40526567386173
8 4 -19.411798038034174
8 8 996.3609940611117
8 12 893.0557750821624
8 16 863.2394210031627
12 0 33.65448439927179 4
12 4 27.85132213963292
12 8 33.28939428294478 4
12 12 994.6167645270792
12 16 866.4961121746067
16 0 12.982629122061 32
16 4 10.568338368325655
16 8 15.394675191195882
16 12 10.145180933976736
16 16 982.0100215534134
```



```
0 0 999.9709762039485
0 4 994.3155541857245
0 8 980.2314334019965
0 12 959.5325925573[illegible]
0 16 914.6[illegible]
4 0 1.5[illegible]0098488245493
[illegible] 999.869504019024
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4 12 887.3150852531244
4 16 872.7177949130992
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8 4 -20.781148315356557
8 8 [illegible]97.9816111803746
8 [illegible] 964.7610242139875
8 16 939.2073[illegible]0364193
[illegible] 0 33.157830[illegible]7446691
[illegible] 4 23.36173[illegible]
[illegible] 8 31.859860[illegible]97542377
[illegible] 12 999.9463[illegible]49188315
12 [illegible] 949.8658581974411
16 0 15.[illegible]
16 4 11.6055575651943 1
16 8 13.595319040076518
[illegible] 12 10.477964766301 4
[illegible] 16 997.4998502616959
```
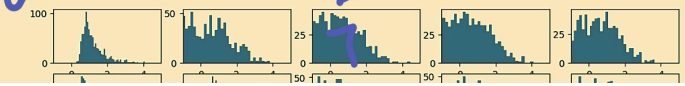



Figures

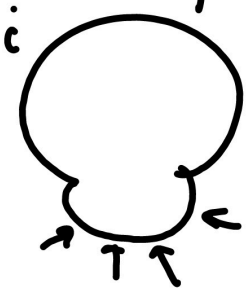
I. one element at a time

1. Passive: apply forces → remove & see shape adapts

a. α - wedging

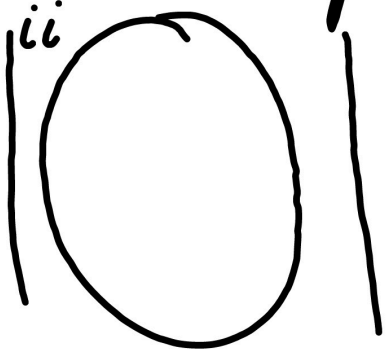
Isotropic α

i

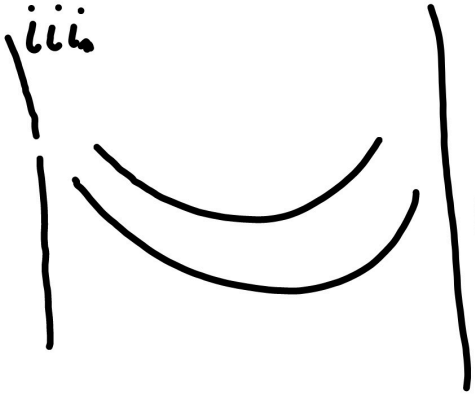


Anisotropic α

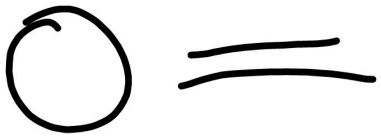
ii



iii.

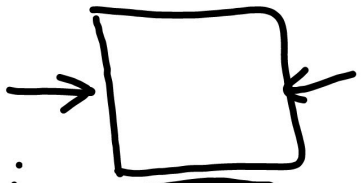


iv. stretch



b. ε - elongation

i

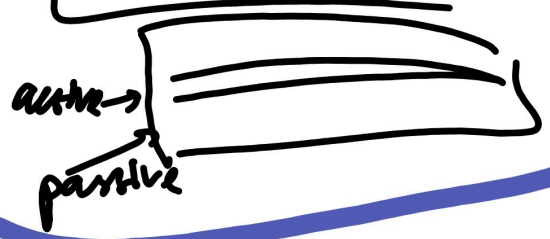


ii

2. Active

a. "Tissue" can elongate/bend from a few local active leaders among a sea of "adaptive"/passive followers

Drosophila example



II. All Combined

Still works!!!

Repeat figs above with combined effects